Constant BOURDREZ

SUMMARY

PhD candidate in Machine Learning at ENS-PSL, working on generative models and privacy, with an engineering background from ESPCI Paris-PSL in physics and data science.

EDUCATION

2025–Present, Paris, France
Ecole Normale Supérieure-PSL, *PhD Candidate in Machine Learning*
PhD Candidate in Computer Science and Machine Learning, Computer Science department and Data Science center at ENS-PSL.

2024–2025, Paris, France
PSL University, *Msc in Computer Science (IASD)*
Joint-Master of PSL Research University (Paris-Dauphine — ENS Ulm — Mines ParisTech). Key courses: Reinforcement Learning, Optimal Transport, Generative Models, Deep Learning, Large Language Models, Mathematics of Deep Learning, Statistical Physics and Machine Learning.

2021–2024, Paris, France
ESPCI Paris-PSL, *MEng in Physics and Data Science*
Research-oriented engineering degree specializing in Physics and Computer Science. Part of the PSL ecosystem, with a transdisciplinary curriculum bridging fundamental research and real-world applications.

2019–2021, Paris, France
Lycée Henri-IV, Paris, France, *CPGE PCSI/PC**
Two-year intensive programme at the university level in mathematics, physics and chemistry, preparing students for the highly competitive nationwide engineering school admissions exam.

RESEARCH AND WORK EXPERIENCE

2025 – Present, Paris, France
Ecole Normale Supérieure-PSL, *PhD Candidate*
- Working on Differential Privacy and Generative Models, especially Diffusion Models.

March – September 2025, Paris, France
Ecole Normale Supérieure-PSL, *Research Intern in Machine Learning*
- Worked on inverse reinforcement learning for diffusion models, supervised by Olivier Cappé and Alexandre Vérine.
- Preprint: `https://arxiv.org/abs/2602.08689`

May – August 2024, HCMC, Vietnam
Oxford University, *Research Intern in Machine Learning*
- Project: 'Probabilistic Forecasting of Dengue Fever in HCMC', supervised by Dr. Marc Choisy at OUCRU.
- Developed a graph-based diffusion model using climate data to provide 30-day dengue fever forecasts.

July – December 2023, Paris, France
Deepki, *R&D Data Scientist - Energy Engineer Intern*
- Internship supervised by Dr. Thimothée Thiery.
- Trained, validated and deployed a hierarchical Bayesian model for a semi-physical model, studying climate change impact on building consumption via inverse modelling, MCMC and climate models.

2021–Present, Paris, France
Lycée Henri-IV / Lycée Fénelon / IPESUP, *Oral Examiner*
Conducted weekly examinations for undergraduate students enrolled in a selective mathematics programme, with the objective of preparing them for engineering schools.

July 2022, Paris, France
Institut Langevin, *Research Intern in Wave Physics*
- Studied Faraday instability in a random environment, supervised by Dr. Emmanuel Fort; reconstructed surface waves via image and signal processing.

ACHIEVEMENTS

Paris Artificial Intelligence Research Institute Excellence Scholarship (Pr[Ai]rie)
3IA Pr[Ai]rie scholarship for the academic year 2024-2025 for the Ms-IASD programme, awarded on academic criteria.

SKILLS AND INTERESTS

Computer Skills: Advanced Python (Numpyro, Xarray, Pytorch, Numpy, Seaborn, Matplotlib, Sklearn, Streamlit, Pandas, Pytorch Lightning, Pytorch Geometric, JAX), Advanced C++, Matlab and Latex. Intermediate Microsoft office and SQL. Git, Docker, GCP

Language Skills: Fluent in English (Cambridge C1 certificate obtained in 2019 and 990 TOEIC score), French native speaker, intermediate Spanish.

Interests: Cycling (Fixed gear and road), running, electronic music, volley-ball and drums.

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